

Laboratory Work #2

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NORMALIZATION

1. First Normal Form (1NF) A table is in 1NF if all fields contain single atomic values and there are no repeated groups.
 - Status: Satisfied.
 - Reason: All attributes in the tables store atomic values. Each table has a defined primary key.
2. Second Normal Form (2NF) A table is in 2NF if it is in 1NF and all non-key attributes depend on the entire primary key.
 - Status: Satisfied.
 - Reason: All primary keys consist of a single column (e.g., flight_id, passenger_id). There are no composite primary keys, so partial dependencies do not exist.
3. Third Normal Form (3NF) A table is in 3NF if it is in 2NF and has no transitive dependencies (non-key columns depending on other non-key columns).
 - Status: Satisfied.
 - Reason: Non-key columns directly depend only on the primary key. All foreign entity references use foreign keys.

DDL

1. Initial Table Creation

```
CREATE TABLE Airline_info ( airline_id INT PRIMARY KEY, airline_code  
VARCHAR(30) NOT NULL, airline_name VARCHAR(50) NOT NULL, airline_country  
VARCHAR(50) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at  
TIMESTAMP NOT NULL, info VARCHAR(50) NOT NULL );
```

```
CREATE TABLE Airport ( airport_id INT PRIMARY KEY, airport_name VARCHAR(50)  
NOT NULL, country VARCHAR(50) NOT NULL, state VARCHAR(50) NOT NULL, city  
VARCHAR(50) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at  
TIMESTAMP NOT NULL );
```

```
CREATE TABLE Passengers ( passenger_id INT PRIMARY KEY, first_name  
VARCHAR(50) NOT NULL, last_name VARCHAR(50) NOT NULL, date_of_birth DATE  
NOT NULL, gender VARCHAR(50) NOT NULL, country_of_citizenship VARCHAR(50)  
NOT NULL, country_of_residence VARCHAR(50) NOT NULL, passport_number  
VARCHAR(20) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at  
TIMESTAMP NOT NULL );
```

```
CREATE TABLE Flights ( flight_id INT PRIMARY KEY, sch_departure_time  
TIMESTAMP NOT NULL, sch_arrival_time TIMESTAMP NOT NULL,  
departing_airport_id INT NOT NULL, arriving_airport_id INT NOT NULL, departing_gate  
VARCHAR(50) NOT NULL, arriving_gate VARCHAR(50) NOT NULL, airline_id INT  
NOT NULL, act_departure_time TIMESTAMP NOT NULL, act_arrival_time TIMESTAMP  
NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL  
);
```

```
CREATE TABLE Booking ( booking_id INT PRIMARY KEY, flight_id INT NOT NULL,  
passenger_id INT NOT NULL, booking_platform VARCHAR(50) NOT NULL, created_at
```

TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL, status VARCHAR(50) NOT NULL, price DECIMAL(7,2) NOT NULL);

CREATE TABLE Booking_flight (booking_flight_id INT PRIMARY KEY, booking_id INT NOT NULL, flight_id INT NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL);

CREATE TABLE Boarding_pass (boarding_pass_id INT PRIMARY KEY, booking_id INT NOT NULL, seat VARCHAR(50) NOT NULL, boarding_time TIMESTAMP NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL);

CREATE TABLE Baggage (baggage_id INT PRIMARY KEY, weight_in_kg DECIMAL(4,2) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL, booking_id INT NOT NULL);

CREATE TABLE Baggage_check (baggage_check_id INT PRIMARY KEY, check_result VARCHAR(50) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL, booking_id INT NOT NULL, passenger_id INT NOT NULL);

CREATE TABLE Security_check (security_check_id INT PRIMARY KEY, check_result VARCHAR(20) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL, passenger_id INT NOT NULL);

2. Schema Alterations (ALTER TABLE)

ALTER TABLE Airline_info RENAME TO Airline;

ALTER TABLE Booking RENAME COLUMN price TO ticket_price;

ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;

ALTER TABLE Airline DROP COLUMN info;

3. Foreign Key Constraints Setup

ALTER TABLE Flights ADD CONSTRAINT fk_flights_departing_airport FOREIGN KEY (departing_airport_id) REFERENCES Airport(airport_id), ADD CONSTRAINT fk_flights_arriving_airport FOREIGN KEY (arriving_airport_id) REFERENCES Airport(airport_id), ADD CONSTRAINT fk_flights_airline FOREIGN KEY (airline_id) REFERENCES Airline(airline_id);

ALTER TABLE Booking ADD CONSTRAINT fk_booking_flight FOREIGN KEY (flight_id) REFERENCES Flights(flight_id), ADD CONSTRAINT fk_booking_passenger FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);

ALTER TABLE Booking_flight ADD CONSTRAINT fk_bf_booking FOREIGN KEY (booking_id) REFERENCES Booking(booking_id), ADD CONSTRAINT fk_bf_flight FOREIGN KEY (flight_id) REFERENCES Flights(flight_id);

ALTER TABLE Boarding_pass ADD CONSTRAINT fk_boarding_pass_booking FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);

ALTER TABLE Baggage ADD CONSTRAINT fk_baggage_booking FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);

```
ALTER TABLE Baggage_check ADD CONSTRAINT fk_baggage_check_booking
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id), ADD CONSTRAINT
fk_baggage_check_passenger FOREIGN KEY (passenger_id) REFERENCES
Passengers(passenger_id);
```

```
ALTER TABLE Security_check ADD CONSTRAINT fk_security_check_passenger
FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
```

4. Sample Data Insertion

```
INSERT INTO airport (airport_id, airport_name, country, state, city, created_at, updated_at)
VALUES (1, 'JFK Airport', 'USA', 'NY', 'New York', NOW(), NOW()), (2, 'Heathrow
Airport', 'UK', 'England', 'London', NOW(), NOW());
```

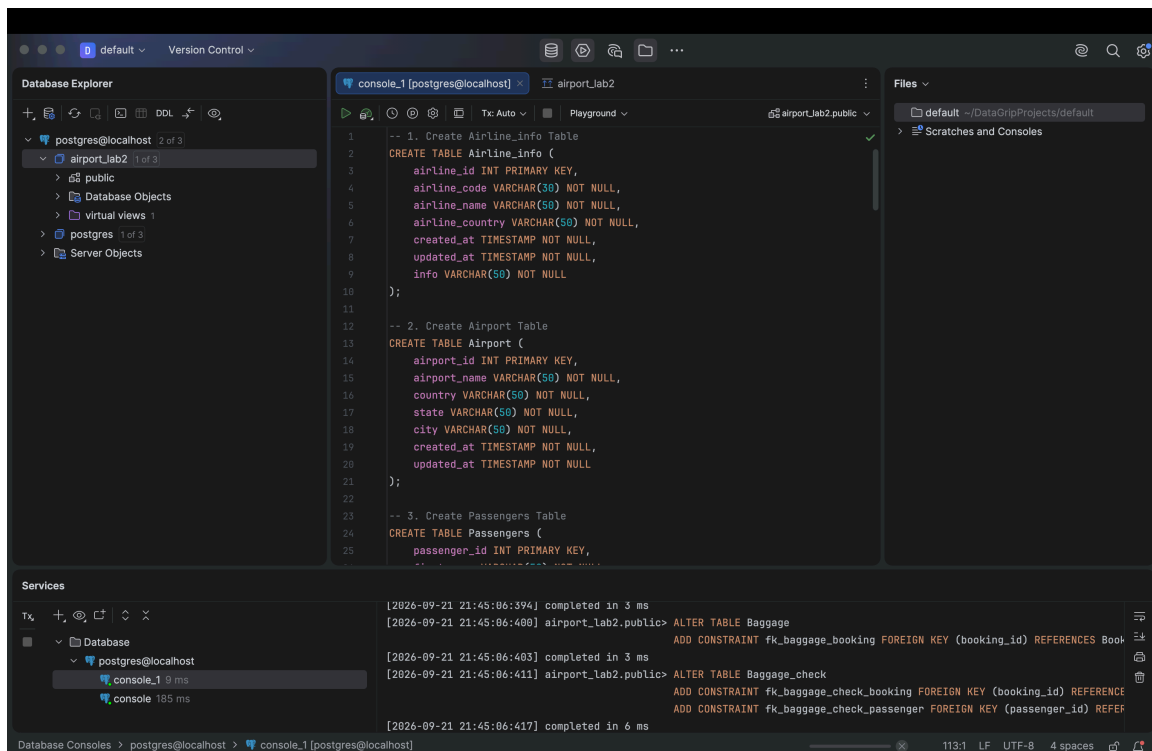
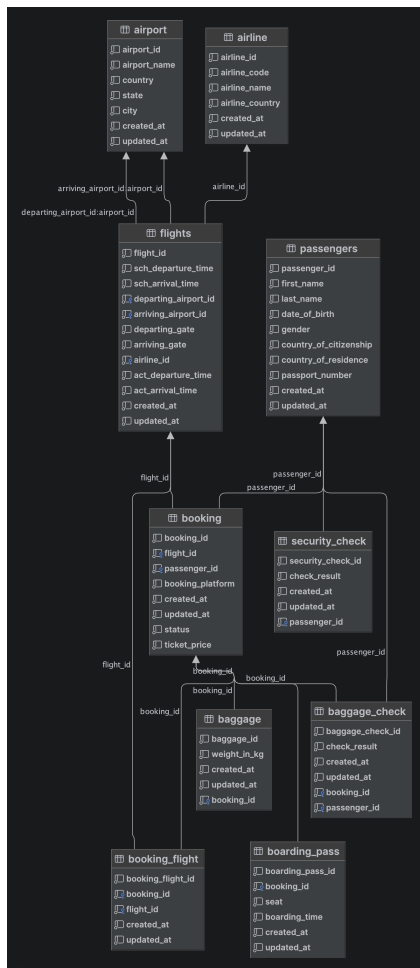
```
INSERT INTO airline (airline_id, airline_code, airline_name, airline_country, created_at,
updated_at) VALUES (1, 'AA', 'American Airlines', 'USA', NOW(), NOW()), (2, 'BA',
'British Airways', 'UK', NOW(), NOW());
```

```
INSERT INTO passengers (passenger_id, first_name, last_name, date_of_birth, gender,
country_of_citizenship, country_of_residence, passport_number, created_at, updated_at)
VALUES (1, 'John', 'Smith', '1995-05-10', 'Male', 'USA', 'USA', 'A12345678', NOW(),
NOW());
```

```
INSERT INTO flights (flight_id, sch_departure_time, sch_arrival_time, departing_airport_id,
arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time,
act_arrival_time, created_at, updated_at) VALUES (101, '2025-10-01 10:00:00', '2025-10-01
18:00:00', 1, 2, 'Gate 12', 'Gate 5', 1, '2025-10-01 10:05:00', '2025-10-01 18:00:00', NOW(),
NOW());
```

```
INSERT INTO booking (booking_id, flight_id, passenger_id, booking_platform, created_at,
updated_at, status, ticket_price) VALUES (501, 101, 1, 'Website', NOW(), NOW(),
'Confirmed', 500.00);
```

```
INSERT INTO boarding_pass (boarding_pass_id, booking_id, seat, boarding_time,
created_at, updated_at) VALUES (701, 501, '14B', '2025-10-01 09:15:00', NOW(), NOW());
```



DataGrip diagrams screenshot